

Yichen (Eason) Guo

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Santa Barbara, CA, USA

EDUCATION

- **University of California, Santa Barbara** Sep 2025 – Jun 2026(expected)
Santa Barbara, CA, USA
Exchange Study in Computer Science
- **Huazhong University of Science and Technology (HUST)** Sep 2022 – July 2026(expected)
Wuhan, China
B.Eng. in Computer Science and Technology
 - Core Courses: Data Structure, Computer Organization, Operating System, Computer Networks, Algorithmic Design & Analysis (90), Database Systems (92), Software Engineering (97), Compiler Principles, Functional Programming (98), AI (96), NLP (94), Big Data (95)

PUBLICATIONS & PREPRINTS

- **OmniTrace: A Unified Framework for Generation-Time Attribution in Omni-Modal LLMs**
Qianqi Yan, Yichen Guo, Ching-Chen Kuo, Shan Jiang, Hang Yin, Yang Zhao, Xin Eric Wang
Under review at the European Conference on Computer Vision (ECCV), 2026. [\[arXiv\]](#)

INTERNSHIP EXPERIENCE

- **ByteDance** Dec 2024 – May 2025
Beijing, China
Software Engineer (full-time, 50hr/wk)
 - Developed an in-house web IDE based on [OpenSumi \(3.6k+ ★\)](#) to replace the legacy front-end editor, which served 1,800+ ML engineers
 - Resolved a core type-checking bug in the OpenSumi debug module ([PR #4450](#)) that caused runtime behavior to differ from VS Code; the fix was officially merged and released in [v3.9.0](#), achieving behavioral parity with VS Code
 - Designed a Node.js monitor service to track CPU, GPU, and disk usage; applied multi-point sampling and mean filtering strategy to improve CPU utilization measurement accuracy by 55.3%, enhancing real-time feedback within IDE containers
 - Modified OpenSumi terminal modules to enable real-time and incremental log display; reduced latency by more than 70% during model compilation and debugging

RESEARCH EXPERIENCE

- **UCSB ERIC Lab, UCSB NLP Group, University of California, Santa Barbara** Oct 2025 – Present
Research Intern
Supervisor: Prof. Xin (Eric) Wang
 - Built a Playwright-powered DOM grounding expert atop [Agent-S3 \(10.9k+ ★\)](#)'s Mixture-of-Grounding stack to replace the UI-TARS visual pipeline; boosted end-to-end interaction success rate from 12.5% to 100% while slashing latency by $\sim 210\times$ ($5.7s \rightarrow 27ms$) to achieve real-time execution
 - Engineered a unified multimodal pipeline utilizing Qwen2.5-Omni-7B and MiniCPM-o-4_5 to execute complex summarization and QA reasoning tasks across the large-scale [MISP-Meeting](#) and [MMAU](#) datasets
 - Architected a novel attention-based source attribution algorithm that maps generated responses to precise temporal ranges, achieving zero-shot multimodal grounding without relying on external embedding models
- **Darko Marinov's Research Group, University of Illinois Urbana-Champaign** Jul 2024 – Aug 2024
Research Intern
Supervisors: Prof. Darko Marinov, Prof. Reyhaneh Jabbarvand
 - **Contributed experimental results to the paper: [Revisiting Test-Case Prioritization on Long-Running Test Suites \(ISSTA 2024\)](#)**
 - Architected and implemented a statistical significance testing module in pure Python, achieving 3.1x greater scalability (81+ comparison groups) over standard R implementations (26-group limit)
 - Optimized the analysis pipeline by eliminating cross-language dependencies, reducing the Docker image size by 45% (200MB) and enabling analysis of 59 TCP techniques across 57K+ test suite runs.
 - Developed a novel Compact Letter Display algorithm and the precise Tukey-Kramer adjustment to automatically categorize techniques by significance ($p < 0.05$), directly validating the paper's central finding.

SELECTED PROJECTS

- **AI-powered Full-stack PCB Defect Detection System**
 - Developed an end-to-end PCB defect detection system integrating YOLOv9, Huawei ModelArts, and a Flask backend; achieved 46.05% mAP@50-95 on 5 defect categories, enabling real-time mobile detection via HarmonyOS
 - Designed a dual-inference architecture combining cloud (ModelArts API) and local (YOLOv9) engines, boosting system availability by 38% under unstable networks through automatic failover
 - Built a low-latency video streaming module using FFmpeg and WebSocket to achieve 360ms local delay and 99.03% uptime; converted RTSP feeds to FLV enhanced real-time visual monitoring for production-line quality control
 - Awarded top national recognition among 400,000+ entries; validated by Huagong Laser and endorsed by Distinguished Professor Jin Hai (Chang Jiang Scholar)

Aug 2024 – Nov 2024

19th National Challenge Cup for College Students - Huawei Cloud Track


- **Unique Studio HR Platform**
 - Architected a recruitment management system serving 600+ users to optimize hiring processes for the technology teams; successfully facilitated three full recruitment cycles while creating 36+ reusable components and replacing redundant watchers with a centralized event bus (mitt) to boost UI response speed by 15-25%
 - Built a real-time analytics dashboard using ECharts with 5 custom chart types (pie, treemap, etc.) to visualize applicant distribution and funnel progression, enabling data-driven hiring decisions
 - Modeled the 8-step recruitment process as a client-side state machine; developed a dynamic notification system with 6+ templates, and managed complex multi-team interview scheduling

Mar 2024 – May 2024

Individual Project at Unique Studio



TECHNICAL SKILLS

- **Programming Languages:** TypeScript, JavaScript, Python, C/C++, LaTeX, ArkTS
- **Frameworks:** PyTorch, React.js, Vue.js, Next.js, Flask, Django
- **Development Tools & Platforms:** Linux, Git, Docker, Kubernetes, OpenSumi

HONORS AND AWARDS

- **National Scholarship, Second Prize**
 - (Award Rate: Top 4%) Ministry of Education, China
 - **National Scholarship, Third Prize**
 - (Award Rate: Top 9%) Ministry of Education, China
 - **National Second Prize**
 - (Award Rate: Top 3.1%) 6th Global Campus AI Algorithm Elite Competition
 - **National Grand Prize**
 - (Award Rate: Top 0.03%) 19th National Challenge Cup - Huawei Cloud Track

Oct 2025

Dec 2024

Dec 2024

Nov 2024